T.C. ALTINBAŞ UNIVERSITY
Software Engineering Department

Furniture E-Commerce Mobile Application: thing.

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Course: SWE332 Software Architecture
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Software Project Proposal Draft#1

1. What will your software do?

thing. is a native mobile e-commerce application designed for buying and selling high-quality furniture products such as cupboards, chairs, tables, and home accessories. The application connects users who are looking for modern furniture with a seamless digital shopping experience. Users can browse categories, view detailed product information, create accounts, and place orders through a clean and user-friendly mobile interface.

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2. What features will it have?

The application will include the following core functionalities and technical foundations:

Core User Features

- **User Authentication:** Secure registration and login via Email/Password and Google utilizing Firebase Authentication.
- **Product Discovery:** Categorized product listing, search functionality, and detailed product pages.
- **Shopping Cart & Orders:** Full "Add to Cart" functionality, quantity management, and a secure order placement system.
- **Profile Management:** User profile editing and order history tracking.

Technical Features & Architecture

- **MVVM Architecture:** Clean and maintainable code structure separating UI logic from business logic.
- **Dependency Injection (Hilt):** Managing dependencies to improve code modularity, scalability, and testability.
- **Real-time Database:** Synchronized data updates using Firebase Firestore.
- **Cloud Storage:** Hosting product and profile images via Firebase Storage.
- **Navigation Component:** A modern Single Activity approach for smooth fragment transitions.
- **Kotlin Coroutines:** Efficient background operations to ensure a lag-free UI experience.

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3. How will it be executed?

The project will be executed as a native Android mobile application developed in Android Studio using the Kotlin programming language.

- **Architecture:** It follows the MVVM (Model-View-ViewModel) architecture pattern to separate business logic from the UI.

- **Backend:** It utilizes Firebase serverless services (Authentication, Firestore, Storage) as the backend solution.
- **Dependency Management:** Hilt will be used for dependency injection to simplify class dependencies and improve scalability.
- **UI Implementation:** The interface will be built using XML and View Binding for type-safe interaction with views.
- **Distribution:** The final application will be compiled and distributed as an APK file.

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4. What new skills will you need to acquire?

To successfully deliver this project, the team will acquire and refine the following skills:

- **Advanced MVVM Implementation:** Mastering the separation of concerns and ViewModel usage.
- **Firebase Security & Management:** Understanding Firebase security rules and efficient Firestore database structuring.
- **Asynchronous Programming:** Managing background tasks and network calls using Kotlin Coroutines and Flows.
- **Performance Optimization:** Techniques for image caching and memory management to ensure a smooth mobile experience.

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5. What topics will you need to research?

- **Mobile E-commerce UX:** Best practices for shopping cart flows and product display layouts.
- **Database Optimization:** Researching Firestore query patterns to reduce costs and increase speed.
- **Data Modeling:** Designing a scalable schema for Users, Products, and Orders in a NoSQL environment.
- **Secure Authentication:** Handling user sessions and password recovery flows securely.

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6. In your team, who will do what?

The team is divided into Backend/Data and Frontend/UI clusters to ensure efficiency:

- **Backend & Data Management (Hasan Açıkel & Doğukan Süme):**
  - Responsible for Firebase Authentication setup, User Profile data management, and implementing Push Notifications.
  - In charge of Firestore Database structuring (Products/Orders schema), creating the Repository layer, and handling Cart logic calculations.

- **Frontend & UI Development (Kağan Şahin & Samed Tevin):**
  - Responsible for the Navigation Component setup, Onboarding/Login screens, and Profile UI implementation.
  - Focused on the Product Listing interfaces, Product Detail pages, and the visual design of the Checkout flow using View Binding.
- **Project Management:** The team will use Asana to track tasks, bugs, and development progress.

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7. What might you consider to be a good outcome for your project? A better outcome? The best outcome?

- **Good Outcome (MVP):** A functional app with a working authentication system, product listing, and basic ordering functionality with a clean mobile UI.
- **Better Outcome:** An app with an optimized Firestore structure, smooth navigation without lag, and a well-structured MVVM architecture that is easy to read and test.
- **Best Outcome:** A complete, professional e-commerce flow (browse → cart → order) with scalable code, crash-free performance, and a highly polished UI/UX design.
